Project Report: Spaceship Titanic — Passenger Transport Prediction

Kaggle Getting Started Competition · Binary Classification · XGBoost · Final Validation Accuracy: 78.9%

1. Problem Statement

Predict whether a passenger aboard the Spaceship Titanic was transported to an alternate dimension following a collision with a spacetime anomaly. This is a supervised binary classification problem, evaluated on classification accuracy.

2. Input / Output Specification

Input (features)

Feature	Type	Description
HomePlanet	Categorical	Earth / Europa / Mars
CryoSleep	Boolean	Whether passenger elected suspended animation for the voyage
Deck / Side	Categorical	Extracted from cabin location string
Destination	Categorical	TRAPPIST-1e / 55 Cancri e / PSO J318.5-22
Age	Numeric	Passenger age in years
VIP	Boolean	Whether passenger paid for VIP service
RoomService, FoodCourt, ShoppingMall, Spa, VRDeck	Numeric	Amount billed per amenity

Output (target)

`Transported` — Boolean (True / False)

3. Data Pipeline Summary

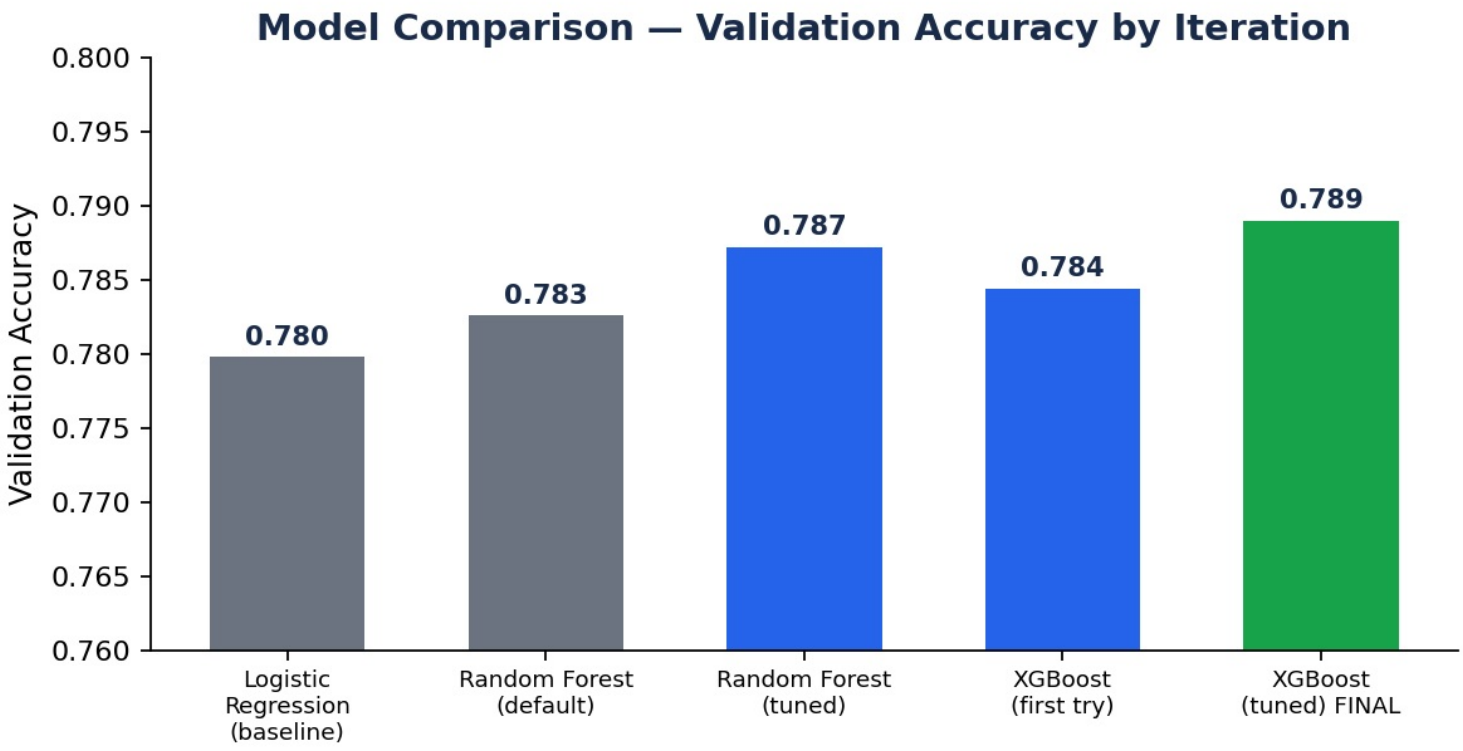
8,693 training rows, ~2% missing values per feature. Missing values were filled using domain logic where possible (e.g. CryoSleep inferred from zero spending, since sleeping passengers cannot spend money; Cabin recovered from known cabins of same-group travelers) and statistical fallback (median / mode) elsewhere, always computed from training data only to avoid leakage into validation/test sets. Categorical features were one-hot encoded; boolean features mapped to 0/1.

4. Model Selection — What Was Tried and Why

Model	Why tried	Val. Accuracy
Logistic Regression	Fast, interpretable baseline — establishes a floor score before investing in complexity	0.7798
Random Forest (default)	Ensemble of trees, expected to outperform a linear baseline on mixed categorical/numeric tabular data	0.7826
Random Forest (tuned)	Default RF barely beat baseline — diagnosed as overfitting; fixed via `max_depth=10`, `min_samples_leaf=5`	0.7872
XGBoost (tuned via RandomizedSearchCV, 3-fold CV)	Gradient boosting typically outperforms bagged trees on structured/tabular data; systematic search over n_estimators/max_depth/learning_rate/subsample/colsample_bytree	**0.7890 (final model)**

Why XGBoost was the final choice

XGBoost's sequential, error-correcting tree structure outperformed both the linear baseline and the bagged-tree Random Forest after systematic hyperparameter tuning with 3-fold cross-validation (cross-validated score: 0.8077, held-out validation: 0.7890 — the gap is expected variance between CV folds and a single held-out split, not a sign of instability). Final hyperparameters: `n_estimators=400, max_depth=6, learning_rate=0.01, subsample=0.8, colsample_bytree=1.0`.



5. Evaluation Methodology

- **80/20 train-validation split**, stratified implicitly by random_state fixing for reproducibility
- **3-fold cross-validation** during hyperparameter search, to avoid overfitting to a single split
- **Accuracy** as the metric, appropriate here since the target classes are balanced (~50.4% / 49.6%) — confirmed before committing to this metric

6. Key Design Decisions

Domain-informed imputation over blind statistics: CryoSleep and Cabin missing values were recovered using logical relationships in the data (spending behavior, group travel patterns) rather than defaulting to mean/mode everywhere — this preserved more real signal than a purely statistical approach would have.

No data leakage: all fill statistics (medians, modes) and the feature scaler were fit exclusively on training data, then applied unchanged to validation and test data.

Baseline-first methodology: establishing Logistic Regression's score before trying ensemble methods made it possible to detect that the first Random Forest attempt was underperforming — a signal that would have been invisible without a baseline to compare against.

7. Result

Final model: **XGBoost Classifier**, tuned via RandomizedSearchCV with 3-fold cross-validation. Held-out validation accuracy: **78.90%**. Predictions generated on the official Kaggle test set and submitted as `submission.csv` in the required `PassengerId, Transported` format.